

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for data processing services, handling approximately **1,500 requests per minute** during peak loads.
- Implemented complex data retrieval and manipulation logic using **Hibernate**, ensuring efficient interaction with relational databases for core application features.
- Designed and executed unit tests using **JUnit** and **Mockito** for critical Java components, achieving over **85% code coverage** on new feature development.
- Contributed to the design and development of mechanical systems for water and wastewater treatment plants, focusing on **pumping station designs** and **treatment plant layouts**.
- Collaborated with process, civil, and E&I; engineers within a multidisciplinary team to plan and deliver integrated mechanical design packages.
- Produced detailed engineering calculations, including **hydraulic analysis** for gravity and pumped systems, and performed **pump selection** based on operational requirements.
- Containerized **Java Spring Boot** applications using **Docker** for consistent deployment across development and production environments on **AWS EC2**.

SKILLS

Engineering Software – SUMO, WAVE, AFT, Infoworks, Hades, AutoCAD Plant 3D

Core Engineering – Mechanical Design, Process Engineering, Civil Engineering, E&I; Engineering, Hydraulics, Pumping Systems, Treatment Plants

Programming Languages – Java, Python, SQL

Frameworks & Libraries – Spring Boot, Hibernate, JPA, JUnit, Mockito, pandas, NumPy

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Project Management – Budget Management, Schedule Management, Deliverables Tracking, Health & Safety Regulations

PROJECTS

Water Sector Mechanical Design Package (HCL Technologies)

- Led the mechanical design phase for a municipal wastewater pumping station upgrade, developing preliminary and detailed design reports, calculations, and drawings.
- Performed hydraulic calculations for the entire pumped system, selecting and specifying new **high-efficiency pumps** to meet increased flow demands and reduce energy consumption by approximately **15%**.
- Collaborated with civil and E&I; teams to integrate mechanical plant selections and P&IDs; into the overall project design, ensuring seamless system functionality.

Automated Reporting and Data Pipeline (HCL Technologies)

- Developed Python scripts to automate the generation of daily operational reports from multiple enterprise systems, reducing manual effort by an estimated **8 hours per week**.
- Built data pipelines using **pandas** and **SQL** to extract, transform, and load data from PostgreSQL and Oracle databases into a central data repository for analysis.
- Implemented scheduled jobs using cron and basic shell scripting on Linux to ensure timely and consistent report delivery to operational teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)